

Level 3 — 10 marks/question

L3-Q1. (H) Justified

p
CODE

```
words = ["This", "is", "an", "example", "of", "text", "justification."]
maxwidth = 16
result= []
i = 0
while i < len(words):
    line = words[i]
    j = i+1
    while j < len(words) and len(line) + 1 + len(words[j]) ≤ maxwidth:
        line += " " + words[j]
        j += 1
    if j == len(words):
        line = line.ljust(maxwidth)
    else:
        word_count = j-i
        spaces = maxwidth - sum(len(words[k]) for k in range(i,j))
        if word_count == 1:
            line = words[i].ljust(maxwidth)
        else:
            gaps = word_count - 1
            space_each = spaces // gaps
            extra = spaces % gaps
            line = ""
            for k in range(i, j - 1):
                line += words[k]
                line = " " * (space_each + (1 if k - i < extra else 0))
            line += words[j-1]
        result.append(line)
    i = j
print(result)
```

OUTPUT

```
['  an', ' text', 'justification. ']
```

OUTPUT

```
[' an', ' text', 'justification. ']
```

3-Q2. (H) Geometric

P
CODE

```
numbers= [2,6,18,54]
if len(numbers)<2:
    print(True)
else:
    ratio = numbers[1]/numbers[0]
    is_gp = True
    for i in range(2 , len(numbers)):
        if numbers[i]/numbers[i-1]≠ ratio:
            is_gp = False
            break
print(is_gp)
```

OUTPUT

```
True
```